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1. A data pipeline is designed to process a large dataset in parallel across hundreds of worker nodes. During a large job, one of these worker nodes unexpectedly fails. According to the principles of resilient distributed processing design, what is the expected outcome?

1/1 point

- ☐ The central coordinator will automatically reassign the failed worker's task to an available node.
- ☐ The entire pipeline job will fail immediately and require a full manual restart.
- ☐ The data assigned to the failed worker will be dropped, leading to an incomplete but successful job.
- ☐ The pipeline will pause and wait for you to manually replace the failed worker.
- ☒ Correct
Correct: Distributed processing systems are designed for fault tolerance. The central coordinator detects the failure and ensures pipeline resilience by automatically reassigning the incomplete task to a healthy worker, allowing the job to finish successfully.

2. You need to combine two separate datasets in a pipeline. The first dataset contains transaction data, and the second contains user profile information. Both share a common user_id field. Which Dataflow transform is designed specifically for joining these two distinct datasets based on their shared key?

1/1 point

- ☐ GroupByKey
- ☒ CoGroupByKey
- ☐ ParDo
- ☐ Combine
- ☒ Correct
Correct: CoGroupByKey is the correct transform for performing a join-like operation. It is designed specifically to group records from two or more datasets together based on a common key, making it the perfect tool for this task.

3. A team's primary architectural goal is to use a single programming model that allows them to define their pipeline logic once and have it run on different execution engines (e.g., Dataflow, Serverless for Apache Spark) in the future. Which Google Cloud service is portable and is built on this "write once, run anywhere" principle?

1/1 point

- ☐ BigQuery
- ☐ Serverless for Apache Spark
- ☒ Dataflow
- ☒ Correct
That's right! Dataflow is the managed execution service for pipelines written with the Apache Beam SDK. Beam is explicitly designed to be a portable, unified model, allowing you to decouple your pipeline's logic from the underlying engine.

4. You are building a new pipeline using Serverless for Apache Spark. The source data is highly structured, with well-defined columns like user_id and purchase_amount. The primary task involves complex filtering and calculations on these columns. According to modern Spark best practices, which core API should you use to represent and manipulate this data?

1/1 point

- ☐ The RDD (Resilient Distributed Dataset) API
- ☒ The DataFrame API
- ☐ The Broadcast API
- ☒ Correct
Correct: The DataFrame API is the modern standard for working with structured data in Spark. It provides a rich set of functions, is highly optimized for performance, and is much easier to use for column-based operations than the lower-level RDD API.